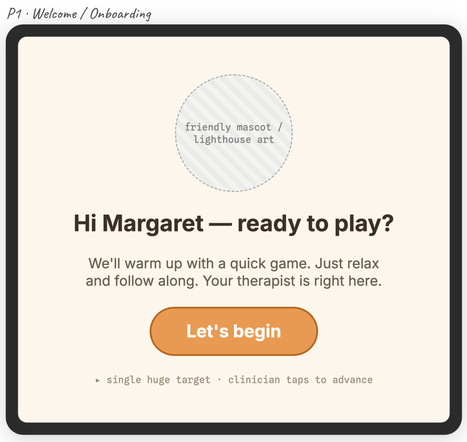
Patient Game App — warm & friendly · hands-free · the brain/movement signal IS the input

P1 · Welcome / Onboarding



DEV · P1 Onboarding scene (Unity)

PURPOSE
Orient patient at session start. One huge affordance, no small targets. Clinician-driven by default; optional large patient button.

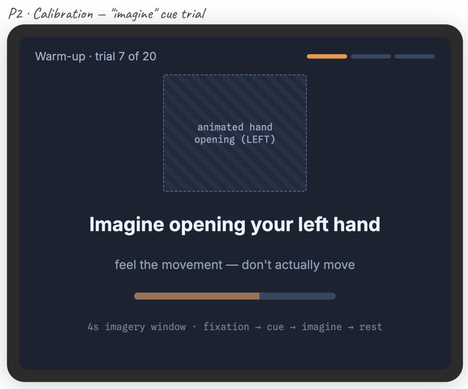
DATA IN
patient.first_name, **affected_side** from profile (C1). No live signal subscription yet.

STATES
Returning (greet by name) vs first-ever → show pictographic explainer + consent/assent reminder. Optional spoken cue (aphasia).

A11Y
Icon-first, short high-contrast large text; primary element pinned to intact visual field; no flicker.

SPEC §9.3 · §9.5 · §12 (consent)

P2 · Calibration — "imagine" cue trial



DEV · P2 Graz cue-paced calibration

PURPOSE
Collect labeled MI trials to fit per-session **CSP + LDA** (or Riemannian). ~20 trials/class.

TRIAL TIMELINE
2s fixation → 1.25s iconic cue → **4s imagery window** → 2–3s randomized rest. Kinesthetic instruction ("feel it, don't move").

DATA IN / OUT
In: marker stream events + live EEG quality; camera invalidates covert trials w/ movement. Out: filters fit offline (<1s).

QUALITY GATE
CV accuracy < 60% → auto-switch to **Blended Mode** + flag on console (handles ~40% "BCI illiteracy").

STATES
fixation / cue / imagine / rest / invalid (movement) / block-complete. Dots = blocks.

SPEC §6 · §6.5 · §9.5

P3 · Game "Garden Grow" — covert MI



DEV · P3 Garden Grow — Mode A (covert MI)

PURPOSE
Lowest-cognitive-load game for covert MI / early-stage. Continuous (not binary) neurofeedback.

MECHANIC
Flower grows with sustained Intent Score; **wilts slightly** on low engagement, recovers — never punitive reset.

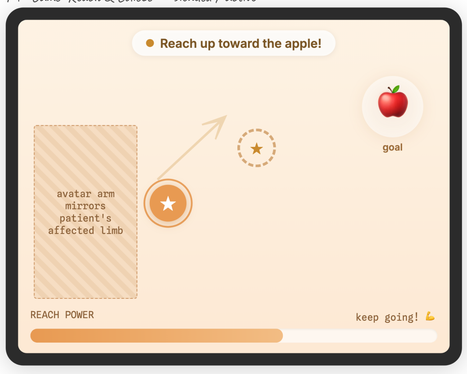
DATA IN
Intent Score 0–100 @ ~20Hz (WebSocket), EMA-smoothed τ=300ms. Mode A: w_eeg=1.0, w_kin=0; **any movement → invalid** (camera gate).

REALTIME
EEG→visible response < 250ms p95. Bloom threshold on up/down staircase → ~70–80% success band.

A11Y / SAFETY
Status pill is plain-language + color (colorblind-safe); no flicker/strobe; no menus (signal is the input).

SPEC §9.1 · §9.2 · §8.2 · §8.5 · §9.4 · §12

P4 · Game "Reach & Collect" — blended / active



DEV · P4 Reach & Collect — Modes B/C (blended & active)

PURPOSE
ROM-targeted game. **Embodied avatar arm mirrors the patient's affected limb** reaching toward a target placed at a clinically-relevant range-of-motion goal.

MECHANIC
Reach extends proportionally to combined Intent Score. **Checkpoints show the reach DIRECTION**; current reach = filled/pulsing ★, next = dashed ★, apple = goal. The big ↗ arrow signals where to reach.

NO DEGREES FOR PATIENT
Angles are deliberately hidden — patient sees direction + glow only. **Actual ROM° is surfaced on the clinician live monitor (C3) & history (C4).**

REACH AXIS = CLINICIAN TOGGLE
Horizontal (reach-out) or vertical (reach-up / shoulder elevation), set per-protocol in **C5**. Avatar + checkpoint path re-orient to match.

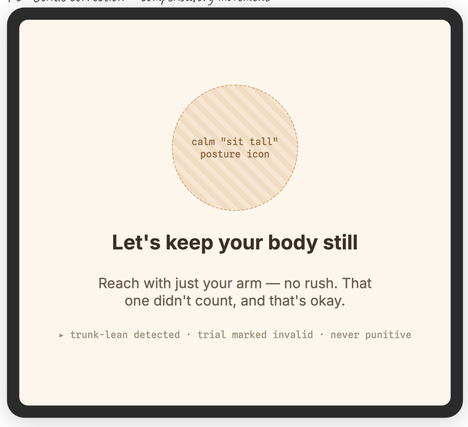
METER
Single fused **"Reach Power"** bar. EEG-vs-kinematic breakdown is intentionally clinician-only (C3) — lower patient cognitive load.

DATA IN
Intent Score + raw sub-scores (EEG_Score, Kinematic_Score) for layered visuals. Kinematic = ROM/velocity toward target. Modes: Blended w_eeg .5–.7 / Active .2–.3.

GUARDRAIL
Trunk-lean → Penalty_Compensation + triggers P5 coaching overlay; reduces (never zeros) score in active mode.

SPEC §9.2 · §9.1 (embodiment) · §7.2 · §8.1 · §8.3–8.4

P5 · Gentle correction — compensatory movement



DEV · P5 Compensation coaching overlay

PURPOSE
Non-punitive overlay when compensatory trunk-lean is detected. Mirrors standard PT compensation-avoidance coaching.

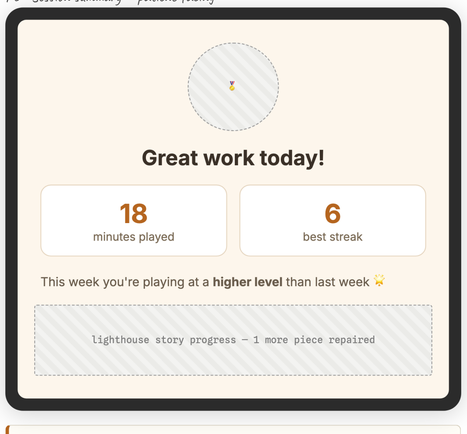
TRIGGER
Trunk displacement > calibrated threshold vs seated rest-pose. Covert trial → invalid + excluded from training; active trial → score reduced + this cue.

DATA IN
Pose stream (trunk displacement), trial valid_flag. Rest-pose from 5s session-start camera calibration.

BEHAVIOR
Transient overlay on the current mini-game (NOT a separate scene); auto-dismiss once posture corrected. Copy always encouraging.

SPEC §7.2 · §8.1 · §9.3 · §16 (frustration risk)

P6 · Session summary — patient-facing



DEV · P6 SessionSummary scene

PURPOSE
Simple, encouraging recap. Rewards **effort, not classifier output** (nobody feels like a "BCI failure").

DATA IN
Session aggregates (duration, best streak, trials); weekly adaptive difficulty level vs last week.

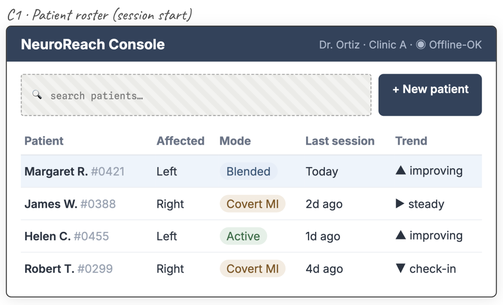
NO CLINICAL METRICS
Patient/kiosk role shows NO PHI or clinical scores. Story Mode (Lighthouse) progress for multi-week adherence.

BUILD
All session events also logged to SQLite for the console + outcome reporting.

SPEC §1 · §9.5 · §9.4 · §11.2 (kiosk role)

Clinician Console — dense & tabbed · EHR + rehab-tracker familiarity · information-forward, nothing buried

C1 · Patient roster (session start)



DEV · C1 Patient roster / worklist

PURPOSE
Session-start patient picker. Mirrors EMR worklist familiarity → low learning curve.

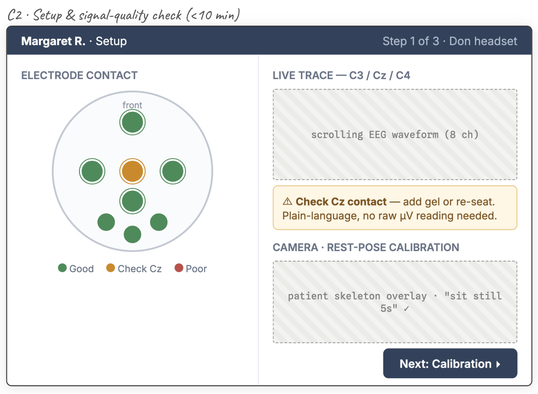
DATA IN
Patient table from SQLite: id, affected_side, last mode, last session date, trend (derived from recent sessions).

INTERACTION
Search; row select → opens C2 setup. "+ New patient" → profile form (affected side, severity, contraindications, consent field).

ACCESS / OFFLINE
Login required; clinician role = full access (kiosk role sees no PHI). Local-first; "Offline-OK" badge.

SPEC §11.1 · §11.2 · §13 · §14

C2 · Setup & signal-quality check (<10 min)



DEV · C2 Setup & signal-quality

PURPOSE
<10-min setup (hard product metric). Raw EEG quality is **translated to plain language** — clinician never reads µV.

STEPS
Don headset (montage Fz, C3, Cz, C4, Pz, PO7, Oz, PO8) → contact map → 5s camera rest-pose calibration → calibration block.

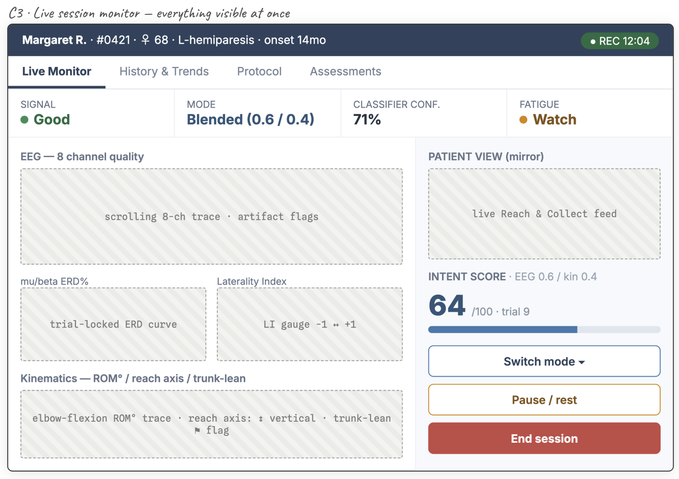
DATA IN
Per-channel quality/impedance over LSL (reuse Unicorn quality scope); pose stream for rest pose.

PLAIN LANGUAGE
"Good signal" / "Check Cz contact" / "Poor". C3/Cz/C4 = MI-primary; others = artifact/context channels.

STATES
per-electrode good/check/poor; camera ok vs occluded (confidence-gated landmarks).

SPEC §10.1 · §10.3 · §7.2 · §7.3 · App A

C3 · Live session monitor — everything visible at once



DEV · C3 Live session monitor (hero screen)

PURPOSE
Single-screen live oversight so the clinician can intervene **without interpreting raw EEG**. Nothing buried.

STATUS TILES
Signal (plain language) · mode + weights · classifier confidence · **fatigue proxy** (declining quality+conf+kinematics → "consider ending").

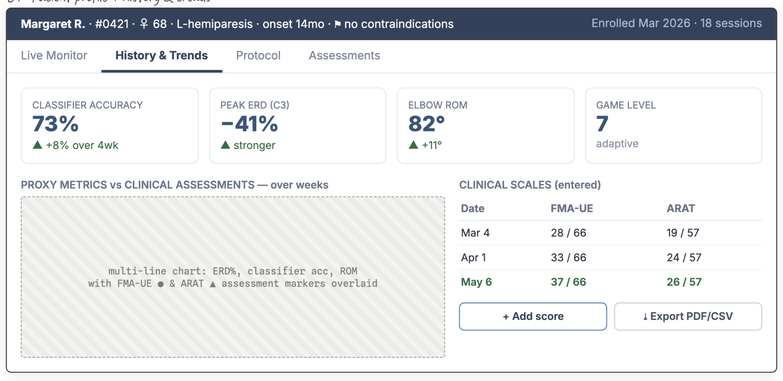
PANELS
8-ch EEG quality + artifact flags · mu/beta ERD% · Laterality Index LI=(ERD_C3−ERD_C4)/(ERD_C3+ERD_C4) · **kinematics ROM° + reach-axis (H/V) + trunk-lean** · patient-view mirror · Intent Score with raw EEG/kinematic sub-scores.

DEGREES LIVE HERE
Actual ROM° (hidden from patient in P4) is shown to the clinician here.

CONTROLS
Switch mode · pause/rest · end session anytime. Data: WebSocket score + marker + pose @ ~20Hz.

SPEC §10.3 · §5.3 (LI) · §12 (fatigue) · §16 (override)

C4 · Patient profile + history & trends



DEV · C4 History & trends

PURPOSE
Week-over-week trends; cross-reference system proxy metrics against formal clinical scales.

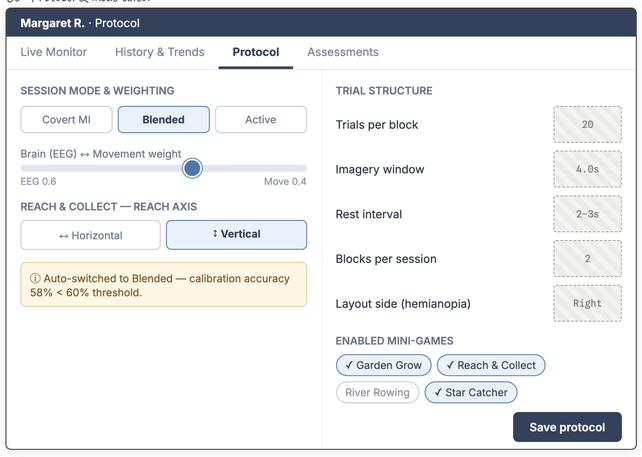
KPIs
Classifier accuracy, peak ERD (C3), elbow ROM°, adaptive game level — all tracked over weeks (incl. degrees hidden from patient).

CHART
Proxy metrics (ERD%, classifier acc, ROM) with **FMA-UE (0–66)** & **ARAT (0–57)** markers overlaid; ARAT MCID ≈ 6.

ENTRY / EXPORT
Clinician enters FMA-UE/ARAT (ClinicalAssessment table). Export: PDF summary, CSV/EDF raw (de-identified for research).

SPEC §1 · §10.3 · §11.1 · §11.2

C5 · Protocol & mode editor



DEV · C5 Protocol & mode editor

PURPOSE
Per-patient protocol config; information-forward (nothing buried under menus).

MODE / WEIGHT
Covert / Blended / Active + EEG↔Movement weight slider. Auto-switch note shown when calibration accuracy < 60%.

REACH AXIS TOGGLE
Horizontal ↔ Vertical for Reach & Collect lives here — drives P4's avatar + checkpoint orientation.

TRIAL STRUCTURE
Trials/block, 4s imagery window, 2–3s rest, blocks/session, + hemianopia layout side.

MINI-GAMES
Enable/disable Garden Grow, Reach & Collect, River Rowing, Star Catcher (one engine, multiple skins).

SPEC §10.3 · §6.5 · §8.2–8.4 · §9.2 · §9.3

Wireframe key

= art / chart / live-feed placeholder · warm = patient app · cool = clinician console

good · check · alert (always paired w/ text — colorblind-safe)

DEV · cards under each frame = developer build notes + spec §refs